The Long-Term Impact of Steel Tariffs on U.S. Manufacturing

Lydia Cox

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Presented by Juncheng Jiang, Wenhao Li, Runhuan Wang, Runyu Zhu
Econ 824: International Trade

Overview

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Cox (2026), pp. 1–4 and 31–33

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Data: steel inputs by industry, from 54,818 requests for exclusion from 2018 tariffs

Cox (2026), pp. 1–4 and 31–33

Roadmap

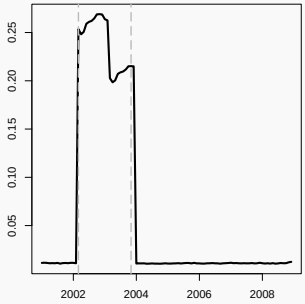
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Bush steel tariffs

A large tariff on steel, removed after almost two years

- Jun 2001: Section 201 probe, 33 steel categories

Import-weighted tariff rate



$\approx 25\%$ in year one, $\approx 20\%$ in year two

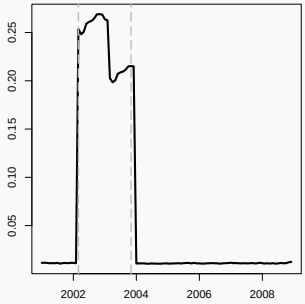
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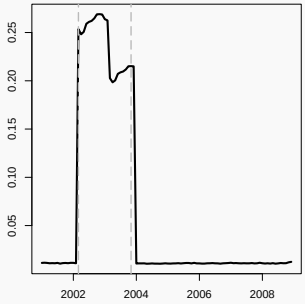
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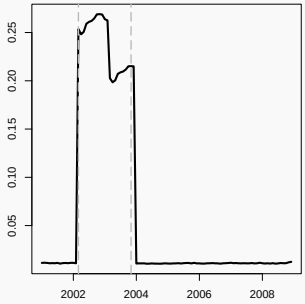
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- Steel users: 80 jobs per steel-making job

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Pass-through to steel users

Foreign suppliers did not cut prices, so U.S. steel users paid the tariff

	Unit value		Import	Import
	excl. tariff	incl. tariff	value	quantity
$\ln(1+\Delta\tau_{sj})$	-0.20	0.37	-3.08	-2.86
	(0.13)	(0.13)	(0.33)	(0.35)

Log change since 2001, HS10 steel product s from country j, 2002–03

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- Producer prices of protected steel rose relative to unprotected steel

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Roadmap

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Model: one buyer, two suppliers

A switching cost can turn a temporary tariff into a permanent loss

- Each period a myopic buyer of good d picks H (U.S.) at p_H or F at $p_H + \varepsilon$, $\varepsilon > 0$
- Changing supplier costs $\bar{\kappa}$: search for a new supplier, customized inputs

Period	Price at H	Price at F	Choice

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Loss persists if $\varepsilon < \bar{\kappa} < \tau - \varepsilon$: tariff beats switching cost, which beats price gap

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Model predictions

Where losses should persist: new relationships, close competitors, costly switching

Cox (2026), pp. 34–36. Recall: switch to F if $\tau > \bar{\kappa} + \varepsilon$, stay at F if $\bar{\kappa} > \varepsilon$

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$\bar{\kappa} = 0$: sharp drop, immediate rebound. Large $\bar{\kappa}$: smaller drop, no rebound.

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Measuring the cost shock

How much did steel tariffs affect each downstream industry's input costs?

- Bush's 2002 tariffs covered only some steel products, with additional rates of 8–30%
- BEA reports total steel spending, but its steel category bundles about 300 HS8 products
- The same steel share can therefore mean different tariff exposure

	Industry A	Industry B
Steel / all intermediate spending	10%	10%
Steel inputs used	Tariffed products	Uncovered products
Average additional steel tariff	30%	0%

We need which steel products each industry uses and their spending shares.

Cox (2026), pp. 12–13. *d*: downstream industry. Numerical comparison is illustrative

Steel-specific input-output data

Exclusion requests from 2018–2019 reveal what each steel product is used for

- The form asks for an **HS10 steel product**, requested kilograms, and its intended use
- For example: “used to make spark plug housings” $\rightarrow$ a downstream automotive use
- Start with **54,818 requests** and **7,187 distinct use descriptions**
- Manually match just over 38,000 requests, about 70%, to downstream uses

Output: detailed steel-to-industry links; requested quantities help construct spending shares next.

Cox (2026), pp. 13–16, Figure 3. Counts refer to requests, not firms. [Coverage]

Steel shares and average tariffs

Index: $d :=$ a downstream industry, $s :=$ a steel product

1. Steel expenditure shares

$$\omega_{s,d} = \frac{\text{Spending}_{s,d}}{\sum_{s'} \text{Spending}_{s',d}}$$

- Numerator: d's spending on steel s
- Denominator: d's spending on all steel products
- $\omega_{s,d}$: this product's share of the steel budget; shares sum to one

2. Average steel tariff increase

$$\Delta\tau_d^{\text{Steel}} = \sum_s \omega_{s,d}\, \Delta\tau_s$$

- $\Delta\tau_s$: additional tariff on steel s
- Multiply each tariff increase by its spending share, then add across products
- Result: d's average tariff increase on its steel inputs

Spending is estimated from requested quantities and 2001 import unit values.

Cox (2026), pp. 16–17 and 20. The weight's denominator includes steel spending only. [Weight checks]

Industry input-cost shock

We need steel's share of spending on all intermediate inputs.

$$\Delta \text{Costs}_d = \underbrace{\ln\left(1 + \Delta\tau_d^{\text{Steel}}\right)}_{\text{steel tariff shock}} \times \underbrace{\alpha_{d,\text{Steel}}^{\text{BEA}}}_{\text{steel spending share}}$$

$$\alpha_{d,\text{Steel}}^{\text{BEA}} = \frac{\text{Steel expenditure}_d}{\text{All intermediate expenditure}_d}$$

- BEA's **2002 Use Table** provides this share; the denominator excludes labor compensation
- For a given tariff increase, a larger steel share means a larger impact on overall input costs
- Result: one **predicted intermediate-cost shock** for each downstream industry

Cox (2026), pp. 12 and 20, eqs. 3–4. Predicted exposure, not observed firm-level total-cost growth

VALIDATING THE COST SHOCK

UMC: material expenditure divided by real shipments, from NBER–CES data for 2001–2003

$$\ln \mathrm{UMC}_{d,t} - \ln \mathrm{UMC}_{d,2001} = a + \beta\, \Delta\mathrm{Costs}_d + \delta_t + \varepsilon_{d,t}$$

- **Outcome**: observed growth in unit material costs in 2002 or 2003, relative to 2001
- **Predictor**: the industry cost shock $\Delta\mathrm{Costs}_d$ constructed above
- **Focus**: does a larger predicted shock correspond to a larger actual cost increase?

Table 6, column (1): $\widehat{\beta} = 1.19$

A 1-percentage-point larger predicted shock is associated with about **1.19 percentage points more UMC growth**.

The positive estimate supports the measure's predictive information.

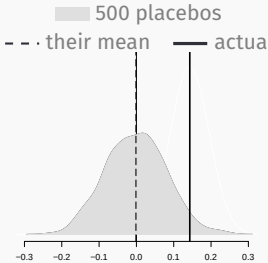
Cox (2026), pp. 20–21, eq. 5 and Table 6 column (1). Percentage-point interpretation is approximate. [2003 survey]

Does the input matching matter?

$$\ln \text{UMC}_{d,t} - \ln \text{UMC}_{d,2001} = a + \beta_H H_d \ln\left(1 + \Delta\tau_d^{\text{Steel}}\right) + \gamma H_d + \delta_t + \varepsilon_{d,t}$$

$H_d = 1$ for industries with **above-median steel intensity**

Estimates of the interaction coefficient



- Would random tariff assignments predict costs just as well?
- Shuffle industry-average tariff changes across industries; keep H_d and UMC fixed
- Re-estimate this equation 500 times
- Placebo estimates center near zero; actual $\widehat{\beta}_H = 0.14$ lies in the right tail

Actual input matching contains information about **material cost changes**.

Cox (2026), pp. 21–22, Table 6 column (2) and Figure 5. Gray density is not a confidence interval. [Steel shares alone]

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Identification

Question: Did the Bush steel tariffs cause larger export losses in more exposed downstream industries?

$$\Delta\tau_d^{\text{Steel}} = \sum_s \omega_{s,d}\,\Delta\tau_s,$$

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- **Endogeneity** : Bush Steel Tariffs
- **Exogeneity** : Variation in Steel Tariff faced by downstream industries (mixture of products)

Specification

Local projections compare industries with larger and smaller cost shocks, year by year

$$y_{d,t} - y_{d,2001} = \alpha_t + \theta_t \, \Delta \text{Costs}_d + \psi_t \, \alpha^{\text{BEA}}_{d,\text{Steel}} + \gamma_t \, \text{Exempt}_{d,2001} + \delta_{h,t} + \varepsilon_{d,t}$$

- θ_t: differential outcome change since 2001 associated with greater predicted cost exposure, estimated separately by year.

Cox (2026), pp. 28–31. $y_{d,t}$: outcome of industry d in year t. [Pre-tariff productivity]

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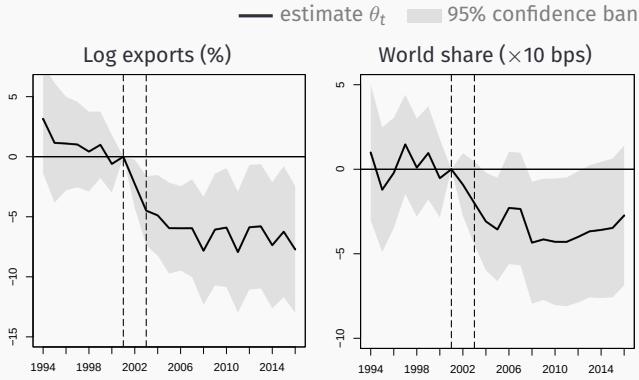
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Credibility

Threat	Why it matters	Evidence / response
Policy selection / lobbying	Protected steel products may be chosen nonrandomly	Policy history; ITC recommendations
Omitted industry trends	Exposed downstream sectors may already be declining	Pre-trends; pre-2002 TFP checks
China shock (2001)	Global competition could be correlated with tariff exposure	NTR-gap control
Exposure measurement	2018–19 input mix may differ from 2002	Alternative and perturbed weights

Exports and world share

Exposed industries lost exports and world share and did not recover after 2003

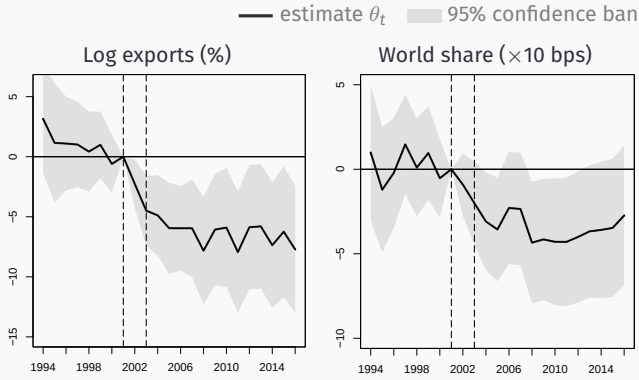


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Cox (2026), pp. 31–32, Figure 7. Responses per 1% cost shock. [Units of share figures]

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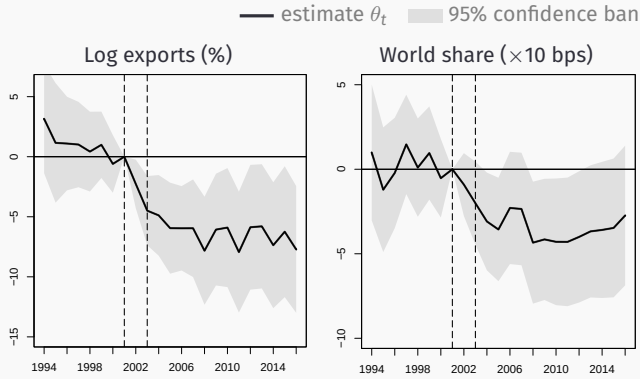


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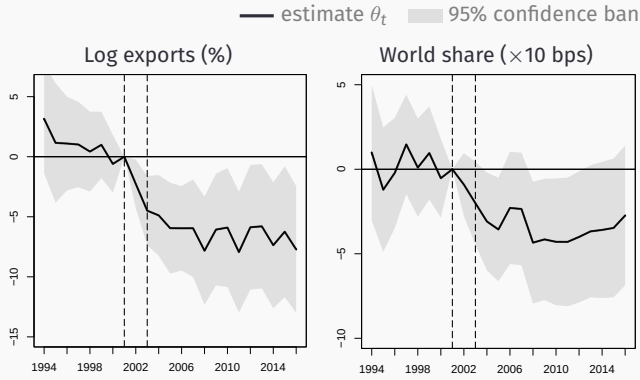


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- **After repeal:** relative loss persists, around -8% in later years.
- **World share:** also declines, with a more gradual response.

Persistence

Proposed mechanism:
Temporary cost shocks may induce foreign buyers to switch suppliers, while the cost of switching back prevents trade relationships from recovering.

Three testable predictions

Cox (2026), Section 5, Figures 9–13.

Do these patterns appear in the data?

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1. **Extensive margin** [Figure]
   Persistence should be associated with fewer active trade relationships.

Cox (2026), Section 5, Figures 9–13.

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   Switching is more likely when foreign suppliers have a small initial price disadvantage.

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3. **Switching costs** [Figure]
   Higher switching costs reduce initial switching, but make switching back less likely.

Cox (2026), Section 5, Figures 9–13.

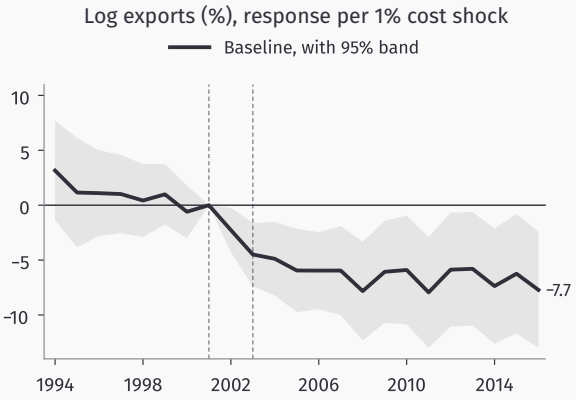
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Roadmap

1. **Setting** Steel products covered by the safeguard
2. **Model** Why a buyer might stay with a new supplier
3. **Measurement** How steel grades determine each industry's exposure
4. **Evidence** How exports evolved after the tariffs expired
5. **Assessment** Alternatives, controls, mechanism, contribution and limitations

Alternative explanations

The 2003 loss survives the main confounders, and persistence is clearest in world share



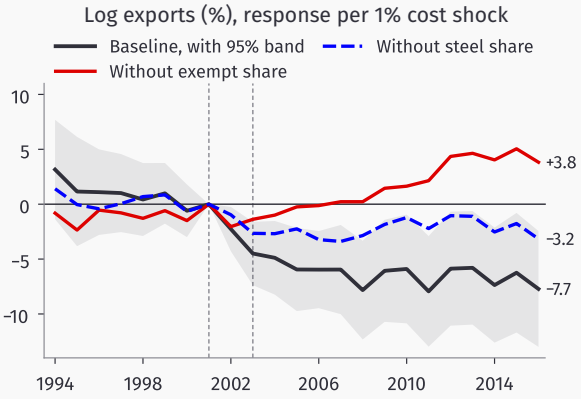
- Lobbying: rates met ITC advice
- Decline: no TFP fall before 2002
- China shock: NTR gap, no change
- Weights: alternatives, no change
- Lake and Liu: job losses by region

2003 stays at -4.5 to -4.6%

Cox (2026), pp. 25–31, Figures 7a and D10, read from vector paths. [Pre-trends] [Lake and Liu] [Prices]

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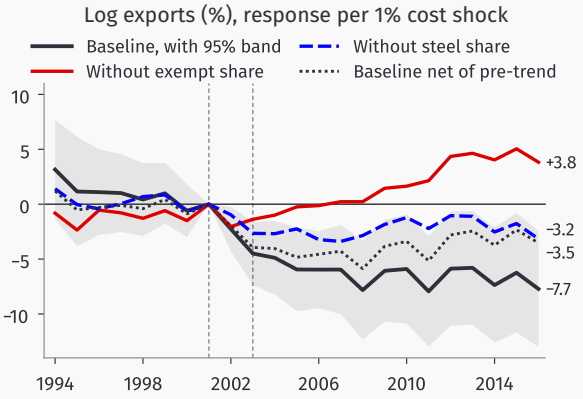


	Mean, 2004–16
Baseline	−6.4
No steel share	−2.3
No exempt share	+2.0

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	Mean, 2004–16
Baseline	-6.4
No steel share	-2.3
No exempt share	$+2.0$
Net of pre-trend	-2.6 to -3.9

World share, net of pre-trend: -22 to -32 bps in all three

Cox (2026), pp. 25–31, Figures 7a and D10, read from vector paths. [Pre-trends] [Lake and Liu] [Prices]

CONTROL VARIABLES

Net of pre-trend, world share is robust to both controls, and exports are not

Control × year	Assessment
Steel share	Steel-heavy products export more after 2008. Without it: world share robust, export loss smaller

Cox (2026), pp. 6, 24, 28 and 59–61, Figure D10. Global Antidumping Database. [Other design checks] [Public regressions]

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Direct tests need Cox's cost shock, which is not public

Cox (2026), pp. 6, 24, 28 and 59–61, Figure D10. Global Antidumping Database. [Other design checks] [Public regressions]

Mechanism: what remains open

Industry data cannot yet isolate buyer switching costs

- **Seller side**: sunk export costs and plant exit also give persistence and fewer relationships [Stories]

Cox (2026), pp. 2 fn. 3, 31 fn. 29 and 37–38, Figures 7b and 10a. Lake and Liu (2025). [Forward-looking buyer]

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A direct test follows the same pre-tariff buyers and asks whether they return [Buyer test]

Cox (2026), pp. 2 fn. 3, 31 fn. 29 and 37–38, Figures 7b and 10a. Lake and Liu (2025). [Forward-looking buyer]

Contribution and limitations

A tariff that lasted under two years lowered exports, and the loss may persist

	Contribution	Limitation
Input tariffs	Handley et al. (2025): lower exports during 2018–19 tariffs. Here, losses last 13 years after repeal	Relative effects: no aggregate effect, no gains to steel producers. Implied elasticity ≥ 6.4, against about 2 [Size]

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Lesson: a temporary tariff's cost includes downstream effects after repeal

Cox (2026), pp. 1–5, 25–27, 36–42, 50 and 58–59. Handley et al. (2025), p. 208. Boehm et al. (2023). Juhász (2018)

Pass-through: full estimates

Imports fell at every level of aggregation, so substitution was partial

Level	Regressor	Unit value		Import	Import
		excl. tariff	incl. tariff	value	quantity
HS10 × country	$\ln(1+\Delta\tau_{sj})$	-0.20	0.37	-3.08	-2.86
		(0.13)	(0.13)	(0.33)	(0.35)
HS10	$\ln(1+\sum_j \omega_{sj}\Delta\tau_{sj})$	-0.03	0.25	-1.22	-1.22
		(0.13)	(0.14)	(0.33)	(0.39)
HS6	HS6 average			-1.04	
				(0.10)	
HS4	HS4 average			-0.63	
				(0.16)	

HS chapters 72–73, 2002–03. ω_{sj}: 2001 import shares. Whole HTS: import value -2.21 (0.20). [Back]

Cox (2026), pp. 7–10 and 46, Tables 1–3 and A1

Steel mix vs. ITC survey

Constructed shares match the ITC survey for cans and parts, less well for vehicles and appliances

Share of steel use (%)	Plate	Hot rolled	Cold rolled	Corrosion resistant	Tin mill	Hot bar	Cold bar	Welded pipe	Stainless bar	Stainless wire	Other
Motor vehicle parts	1	5	8	15	0	16	2	7	6	1	40
ITC, 2002–03	1	6	7	8	1	25	1	1	0	1	50
Motor vehicles	0	3	56	40	0	0	0	2	0	0	0
ITC, 2002–03	0	21	16	72	0	1	0	0	0	0	0
Steel barrels and cans	0	0	14	17	68	0	0	0	0	0	1
ITC, 2002–03	0	0	8	0	83	0	0	0	0	0	9
Household appliances	0	0	14	58	0	0	1	0	0	1	25
ITC, 2002–03	0	2	47	33	1	0	1	0	0	0	17

ITC does not publish its category definitions, e.g. where stainless steel falls. [Back]

Cox (2026), pp. 18–20, Table 5. ITC: U.S. International Trade Commission (2003), Table D-1

Coverage of exclusion requests

Requests cover products worth 95% of 2001 imports of Bush-tariffed steel

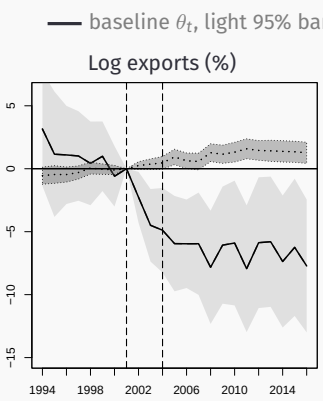
Category	HS8 products	Share of 2001 imports
Full BEA steel product set	273	100%
Excluded upstream steel inputs	5	0.5%
Products covered by 2018 steel tariffs	268	99.5%
Exclusion requests filed (HS8)	88%	96%
Covered at HS6	96%	98%
Covered at HS4	100%	100%
Products covered by Bush steel tariffs	161	–
Exclusion requests filed (HS8)	87%	95%
Covered at HS6	97%	99%
Covered at HS4	100%	100%

Covered at HS6 (HS4): another HS8 product in the same HS6 (HS4) has a request. [Back]

Cox (2026), pp. 25–26, Table 8

Exposure from steel shares alone

Steel shares alone show no export loss



- Regress exports on $\alpha^{\text{BEA}}_{d,\text{Steel}}$ and HTS Section $\times$ year effects only
- Steel share is not exogenous to industry outcomes, so this is a descriptive contrast
- Standard input-output exposure would have missed the export loss

[Back]

Cox (2026), pp. 58–59, Figure D9b

Summary statistics

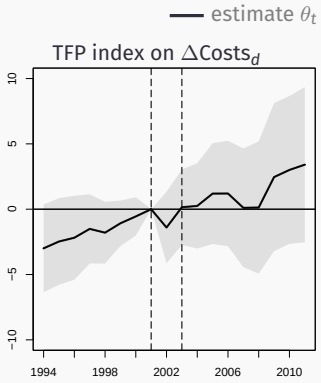
The mean cost shock is 0.2%, far below the 1% unit of the figures

	N	Mean	SD	Min	Max
$\Delta\tau_d^{\text{Steel}}$	5081	0.023	0.071	0.000	0.300
$\alpha_{d,\text{Steel}}^{\text{BEA}}$	4973	0.028	0.052	0.000	0.373
ΔCosts_d	5081	0.002	0.007	0.000	0.098
$\text{Exempt}_{d,2001}$	5081	0.941	0.188	0.000	1.000
U.S. exports (USD million)	5081	131	710	0.008	23464
U.S. share of world exports	5069	0.118	0.127	0.000	0.950

One observation per HS6 industry. [Back]

Productivity of exposed industries

Exposed industries' TFP did not fall before the tariffs



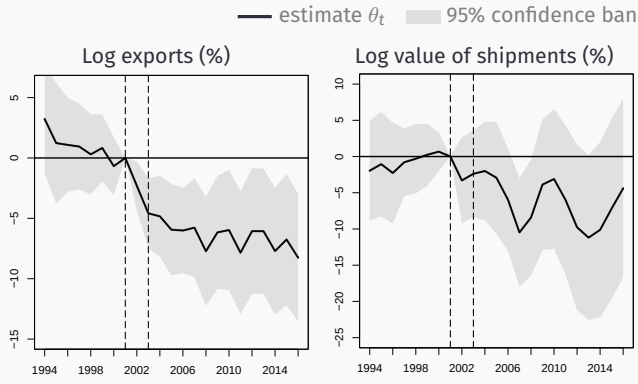
- NBER-CES four-factor TFP index as the outcome of the baseline specification
- No significant relation, before or after 2002. If anything, TFP rises in exposed industries
- Same with $\Delta\tau_d^{\text{Steel}}$ alone and NAICS3 effects, and with TFP growth

[Back]

Cox (2026), pp. 51–53, Figure D2a

Controlling for the China shock

With the NTR gap as a control, the export loss is unchanged



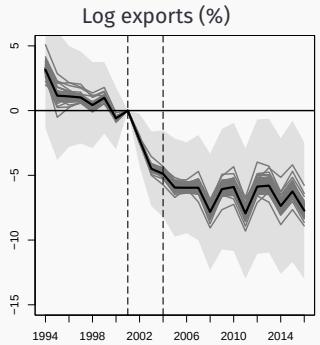
- NTR gap: column 2 minus MFN tariff (Pierce and Schott, 2016)
- Exports: same path as the baseline
- Shipments: insignificant, so the paper drops them
- Employment: imprecise after 2002

[Back]

Cox (2026), pp. 56–58, Figure D8

Robustness of the weights

Dropping any single HS4 sector leaves the export path intact



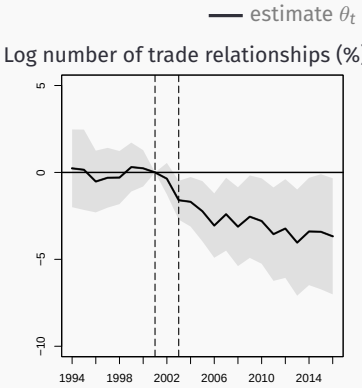
- 100 draws of weights perturbed by up to 5% stay inside the baseline band
- Weights from 2015–17 consumption in place of requested amounts: little change
- Control for substitutability, the gap between 2001 import shares and request shares: little change

[Back]

Cox (2026), pp. 25–27 and 53–57, Figures D3–D5 and D7

Prediction 1: extensive margin

More exposed industries had fewer active trade relationships after repeal



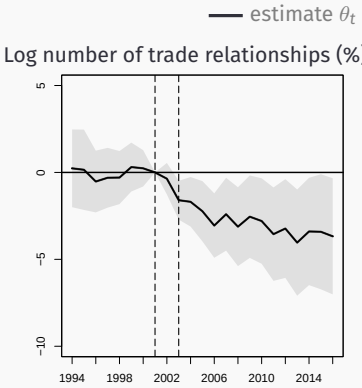
- **Trade relationship**: HS10 product × U.S. customs district × destination, e.g. golf carts from Savannah to Japan

[Back]

Cox (2026), pp. 36–37, Figure 9. Recall: only a new supplier relationship costs $\bar{\kappa}$

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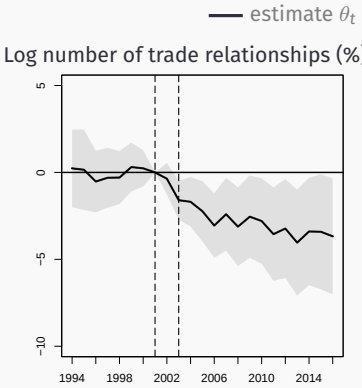
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[Back]

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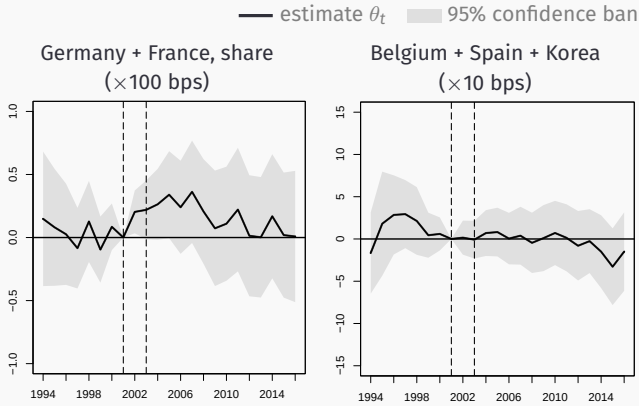
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- Band below 0 every year from 2004
- Only a proxy: buyers and exporting firms are not observed

[Back]

Cox (2026), pp. 36–37, Figure 9. Recall: only a new supplier relationship costs $\bar{\kappa}$

Prediction 2: close competitors

Losses concentrate where the U.S. faced close competitors



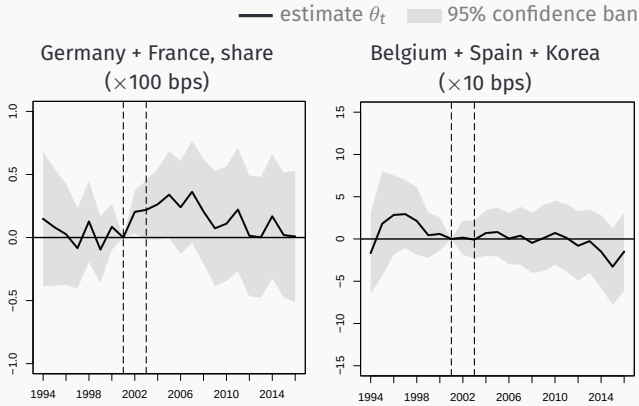
- Top exporters of exposed goods, 2001: Germany, U.S., Japan, France

[Back]

Cox (2026), pp. 37–39, Figures 10–11. Recall: switching needs a small price gap ε

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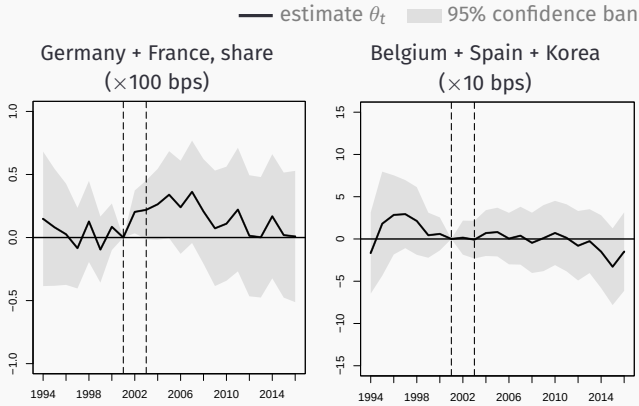
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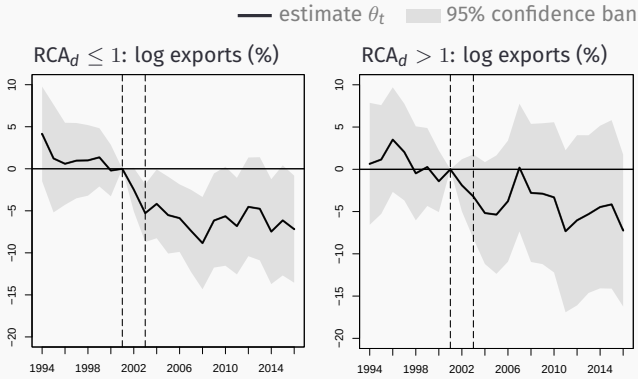
- Top exporters of exposed goods, 2001: Germany, U.S., Japan, France
- Germany and France gain share, 2002–07
- Lower-ranked exporters and Japan: no response [Table]

[Back]

Cox (2026), pp. 37–39, Figures 10–11. Recall: switching needs a small price gap ε

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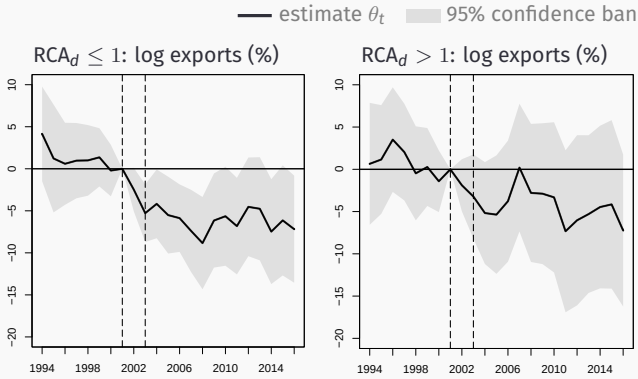
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[Back]

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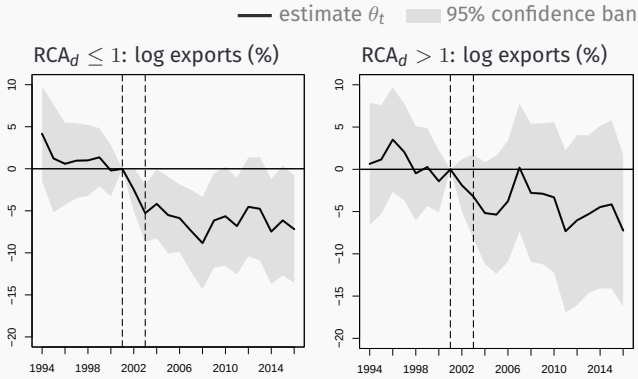
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[Back]

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- RCA$_d > 1$: band includes 0 every year

[Back]

Cox (2026), pp. 37–39, Figures 10–11. Recall: switching needs a small price gap ε

Top exporters of exposed products

Germany, Japan and France were the U.S.'s closest competitors in 2001

	Total exports (USD million)	Average exports (USD million)
Germany	200750	290
USA	172480	249
Japan	157671	228
France	77512	112
Canada	76916	111
Italy	70668	102
UK	51896	75
Belgium	44598	64
Spain	41234	60
China	37834	55
Korea	37710	54

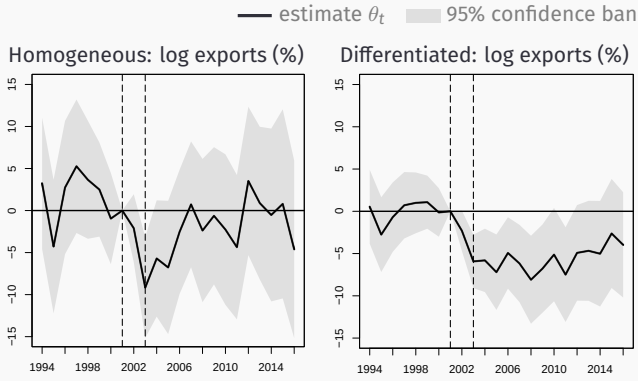
- Exposed products: HS6 goods with a positive tariff on their steel inputs
- Top competitors: Germany and France. Japan's share does not respond
- Lower-ranked: Belgium, Spain and Korea, the bottom three of the top ten excluding China

[Back]

Cox (2026), pp. 38 and 61, Table E1 and Figure E1. UN Comtrade, 2001

Prediction 3: size of switching costs

Differentiated goods show a smaller drop and no rebound



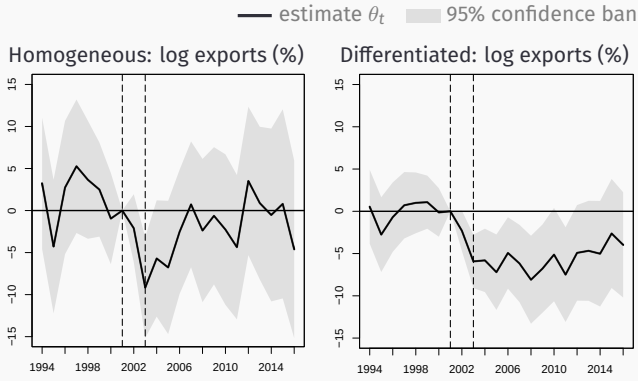
- Rauch (1999): homogeneous goods trade on organized exchanges, so switching is cheap

[Back]

Cox (2026), pp. 39–41, Figures 12–13. Recall: higher $\bar{\kappa}$, fewer buyers leave and fewer return

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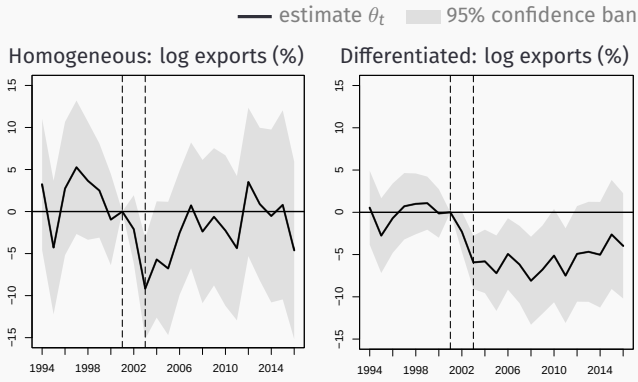
- Rauch (1999): homogeneous goods trade on organized exchanges, so switching is cheap
- Homogeneous: -9% in 2003, near 0 by 2007

[Back]

Cox (2026), pp. 39–41, Figures 12–13. Recall: higher $\bar{\kappa}$, fewer buyers leave and fewer return

Prediction 3: size of switching costs

Differentiated goods show a smaller drop and no rebound



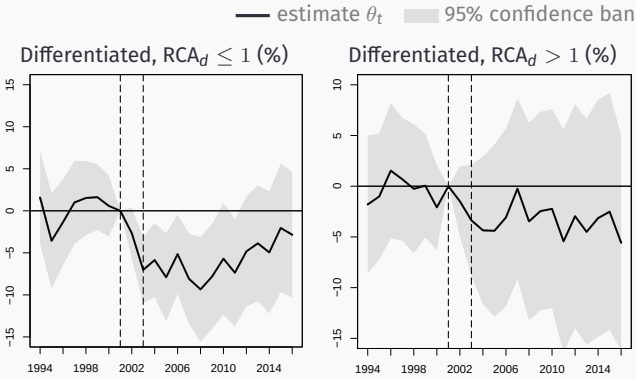
- Rauch (1999): homogeneous goods trade on organized exchanges, so switching is cheap
- Homogeneous: -9% in 2003, near 0 by 2007
- Differentiated: -6% in 2003, -5 to -8% through 2011

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Cox (2026), pp. 39–41, Figures 12–13. Recall: higher $\bar{\kappa}$, fewer buyers leave and fewer return

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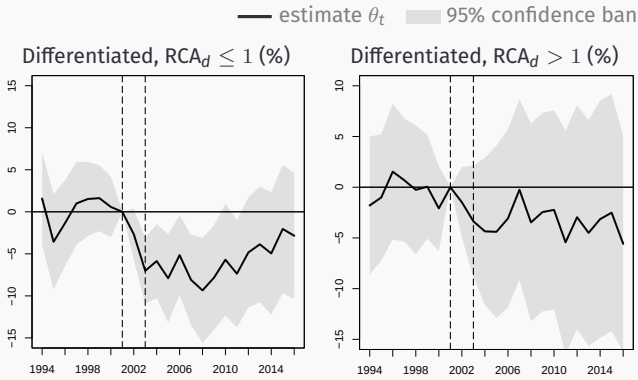
- Differentiated, RCA$_d \leq 1$: -5 to -9% in 2003–11

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Cox (2026), pp. 39–41, Figures 12–13. Recall: higher $\bar{\kappa}$, fewer buyers leave and fewer return

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Differentiated goods show a smaller drop and no rebound



- Differentiated, RCA$_d \leq 1$: -5 to -9% in 2003–11
- Differentiated, RCA$_d > 1$: band includes 0 every year

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Cox (2026), pp. 39–41, Figures 12–13. Recall: higher $\bar{\kappa}$, fewer buyers leave and fewer return

Units of share figures and other inconsistencies

One printed unit in Figure 7b is about 10 bps, which reproduces the text's 6–7%

1. Figure 7b equals 10 times Figures D4a and D5a in every year, for the same baseline
2. Text: trough is "between 6 and 7 percent" of a median share of "roughly 7 percentage points". -43 bps is 6.2%, while -4.3 bps would be 0.6%
3. Exports check: $-6.4\% \times 7$ points $= -45$ bps at median share
4. Inferred from ratios and band widths: 1 unit $\approx$ 10 bps in Figures 7b, 10b, D3a, D7a–D10, E1, and 1 unit $=$ 100 bps in Figures 10a, D4a, D5a
5. Other: differentiated peak "almost half" the homogeneous drop, Figure 12 shows $\approx -8\%$ vs. $\approx -9\%$. Table 3 repeats Table 2's -1.22 with standard error 0.14 for 0.33. Matched requests sum to 102%

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Cox (2026), pp. 1–2, 9–10, 15, 32, 38, 40–41, 54–55 and 60

Pre-trends and test power

Joint tests pass, but they have little power against trends of this size

Outcome	Slope per year	Joint p	80% power	Net of trend
Exports (%)	-0.28, -0.42	0.84, 0.77	0.67, 1.02	-3.9, -2.6
World share (10 bps)	-0.06, -0.05	0.94, 0.74	0.57, 0.83	-3.0, -3.1
Relationships (%)	$+0.02$, $+0.06$	0.99, 0.96	0.32, 0.49	-3.2, -3.6
Employment (%)	$+0.49$, $+0.54$	0.18, 0.44	0.64, 1.13	-11.7, -12.1

- Line through base year, fit to 1994–2000. First value: independent estimates. Second: common base year, correlation 0.5
- Export slope with 80% power gives -6 to -9 at the 2004–16 mean

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Cox (2026), Figures 7, 8a and 9. Rambachan and Roth (2023), smoothness restriction. Roth (2022), pre-test power. Covariances are not published

Comparison with Lake and Liu (2025)

Both study the same tariffs and find downstream harm, using different variation

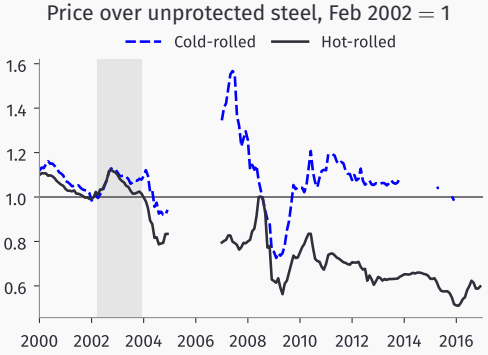
	Cox (2026)	Lake and Liu (2025)
Unit	HS6 and NAICS6 industry	Commuting zone
Years	1994–2016, base 2001	1993–2008, base 2000
Exposure	Tariff on steel mix $\times$ steel share	Tariffs through total requirements
Controls	Steel and exempt shares, Section $\times$ year	Chinese imports, antidumping
Result	Exports lower through 2016	No steel job gain, users' jobs and plants fall
Friction	Buyer switching costs	Plant fixed entry costs

Cox's Figure D9: steel share alone gives no export loss. [Back]

Cox (2026), pp. 28–29 and 59. Lake and Liu (2025), abstract, data appendix and July 2024 draft

Steel prices

Protected steel's premium was gone by 2004, and equal tariffs raised prices unequally



Price paid by users	Tariff	Peak	Mean
Hot-rolled sheet	30%	+76%	+36%
Cold-rolled sheet	30%	+64%	+31%
Galvanized sheet	30%	+57%	+30%
Coiled plate	30%	+27%	+15%
Cold-finished bar	30%	+16%	+10%
Low-carbon wire rod	none	+10%	+4%
Structural beams	none	+10%	+2%
Stainless sheet	none	−3%	−7%

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FRED, replicating Cox (2026), Figure 2a. Blank months: no BLS index. *Purchasing*, via Lake and Liu (2025). Mean: April 2002–September 2003

Regressions on public data

Coefficients on steel share and tariff exposure fall under finer fixed effects

Regressor and outcome	Section × year	HS2 × year	HS4 × year
Steel share: log exports, 2016	$+3.99$ (1.13)	$+2.90$ (1.13)	$+1.23$ (1.13)
Steel share: log quantity, 2016	$+4.24$ (1.20)	$+2.64$ (1.23)	$+1.41$ (1.30)
Steel share: China's share, 2016	$+0.30$ (0.06)	$+0.25$ (0.08)	$+0.21$ (0.16)
Tariff, given steel share: log exports	$+42.3$ (12.9)	$+20.1$ (15.0)	-2.2 (16.6)
Tariff, given steel share: China (bps)	-159 (99)	-81 (122)	-19 (173)

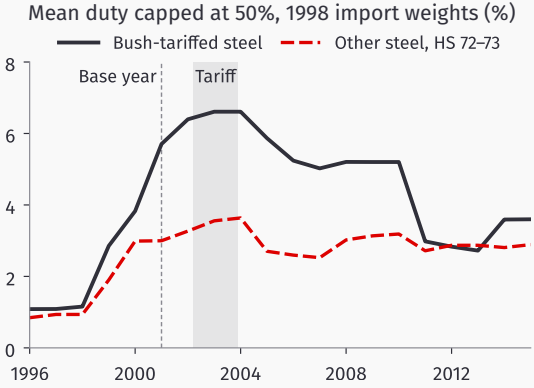
- Exposures come from input–output tables, so these do not re-estimate θ_t
- Steel-share path tracks Cox's Figure D9 in shape (correlation 0.93), and differs in size

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BACI and BEA 1997 Use table, from Lake and Liu's package. Clusters by HS4. Tariff rows: 2004–16 means, upper-bound errors

Antidumping duties

Bush steel's duty gap jumps in Cox's base year and lasts to 2010



- Gap over other steel: $+0.8$ points in 2000, $+3.1$ in 2002–05, $+2.1$ in 2008–10, at most $+0.8$ from 2011
- Relative to 2001: $+0.8$ to $+2.0$ in 2002–05, against a tariff of about 25 and 20
- Summed by year, 2002–10: 4 to 13 points, against 39 for the tariff
- Lake and Liu control for it and still find job losses through 2008

[Back]

Global Antidumping Database, final orders. Other steel: HS 72–73 outside the Bush list. Cox (2026), p. 6. Lake and Liu (2025)

Other design checks

Base year and direct requirements change little

- **Base year**: Section 201 began in 2001, and Lake and Liu use 2000. Rebased 2004–16 mean is -5.8 (2000) or -6.7 (1998–2000), against -6.4
- **Direct requirements**: 170 of 489 input–output industries use no steel directly. Their exposure is 0.08% of costs, against 0.73%
- **Exempt share**: world-share pre-trend is flat with it, about $+2$ bps a year without it
- **Inference**: Cox clusters by HS4. 128 of 171 tariffed HS8 lines pay 30%, so shift-share inference applies (Adão et al. 2019, Borusyak et al. 2022)
- **Weights**: HS6 products count equally. Exports: mean $131M, standard deviation $710M
- **Steel mix of 2018–19**: consumption and perturbed weights keep both controls

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Cox (2026), pp. 24, 28, 32, 50–51, 53–54 and 60–61. Lake and Liu (2025). BEA, from their package

Stories and predictions

Fewer routes fit every story, and no single story fits all five facts

Story	Fewer routes	Rivals keep gains	Plants exit	Homogeneous recover	Export prices
Buyer switching cost (Cox)	yes	yes	no	yes	flat, lower
Exporter sunk cost	yes	yes	no	?	flat
Plant fixed cost (Lake and Liu)	yes	yes	yes	no	flat
Lost scale or learning	yes	yes	?	no	higher
Third-market rival	yes	no	?	no	lower
Facts	yes	no, after 2007	yes	point estimate	flat, lower

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Cox (2026), Figures 7, 9–13 and D1. Lake and Liu (2025). Das et al. (2007). Krugman (1987). Predictions: my reading. Prices: value minus quantity

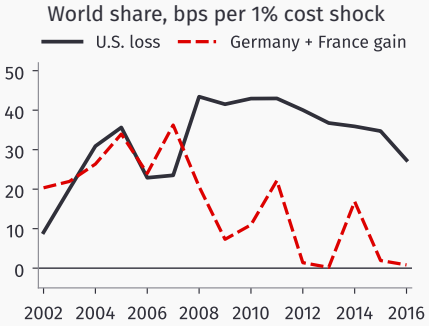
Subsample gaps and rival gains

All gaps have Cox's sign, and one passes 2, only if years are independent

Mean, 2004–16 (%)	Groups		$\lvert z \rvert$
RCA$_d$ ≤ 1, > 1	-6.2	-4.4	0.3–1.2
Homog., differentiated	-1.8	-5.6	0.7–2.6
Diff.: RCA$_d$ ≤ 1, > 1	-5.8	-3.4	0.4–1.4

- U.S. and Germany + France, bps: -28, $+30$ in 2004–07, -43, $+15$ in 2008–11, -35, $+4$ in 2012–16
- $|z|$ runs from upper-bound to independent errors

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Cox (2026), Figures 7b and 10–13, units harmonized on the units slide. Gelman and Stern (2006)

Forward-looking buyer

Cox's condition is the myopic limit of a present-value comparison

- Prices as in Cox's model: tariff adds τ at *H* for *T* years, discount factor β
- Stay at *F* after repeal: $\dfrac{\varepsilon}{1-\beta} < \bar{\kappa}$
- Leave *H* at onset: $\bar{\kappa} + \dfrac{\varepsilon}{1-\beta} < \dfrac{\tau\left(1-\beta^{T}\right)}{1-\beta}$
- Together: $\varepsilon < (1-\beta)\bar{\kappa} < \tau\left(1-\beta^{T}\right) - \varepsilon$. At $\beta = 0$: $\varepsilon < \bar{\kappa} < \tau - \varepsilon$
- Such $\bar{\kappa}$ exists only if $\varepsilon < \tau\left(1-\beta^{T}\right)/2$. At $\beta = 0.95$: 4.9% of τ for $T = 2$, 7.1% for $T = 3$

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Own derivation from Cox (2026), pp. 34–36. Lake and Liu (2025): stakeholders expected three years at most, steel rationed in mid-2002

Size of the export response

The response after repeal implies an elasticity three times a long-run benchmark

	Implied by Cox	Benchmark
2004–16, per 1% shock	-6.4%, elasticity ≥ 6.4	About 2. Net of pre-trend: 2.6–3.9
2008, per 1% shock	-43 bps of world share	Exports imply -45 bps
2002, total	$-\$5.7$B exports	USITC firm survey, scaled: $-\$1.5$B
2004–16, per year	$-\$16$B exports	Net of pre-trend: $-\$7$B to $-\$10$B

- Dollars apply an unweighted relative response to 693 exposed products, $172.5B of exports, mean shock $\approx 1.5\%$
- Temporary shocks should move exports less than permanent ones (Ruhl 2008)

[Back] [USITC welfare]

Cox (2026), pp. 1–2, 32–33 and 50, Tables C1 and E1. Boehm et al. (2023). USITC (2003), Table 2-38. Elasticity assumes pass-through of at most one

Lost customers and plant closures

Two investments can be costly to replace after repeal: a customer relationship and an operating plant.

Evidence on plants Lake and Liu find a lasting decline in the share of steel-intensive manufacturing establishments in more exposed U.S. commuting zones.

Evidence on buyers Cox observes fewer product–customs district–destination routes. Each route can carry orders from several buyers and plants. Existing buyer–supplier pairs carry over 80% of U.S. import value (Monarch and Schmidt-Eisenlohr 2023).

Proposed comparison Follow the same pre-tariff buyers, and measure how often they return when the original plant still offers the product at a competitive delivered price. [Back]

Cox (2026), Figure 9 and p. 31 fn. 29. Lake and Liu (2025), evidence on steel-intensive establishments.

How foreign supply affects the U.S. welfare estimate

Less elastic foreign supply gives a terms-of-trade gain, which raises U.S. welfare

Foreign supply elasticity	5	10 (central)	20
Change in U.S. welfare ($ million)	$+65.6$	-41.6	-110.0

Tariff revenue	Labor	Steel capital	Other gaining capital	Losing capital	GDP
$+649.9$	-386.0	$+239.5$	$+67.4$	-601.2	-30.4

USITC holds capital stocks fixed. Its steel prices rise 0.4–0.9% on a broad steel aggregate, while users' prices rose 30–36% on main flat products. Adding Cox's later losses needs changes in income over time. [Back]

USITC (2003), vol. III, chap. 4, Tables 4-2 and 4-3. Welfare is measured by equivalent variation.